

Homework 3 (Theory) – COP3337

1.

- a. **False** A class definition is not required to contain the keyword `private`. If no access specifier is provided, members of a class are private by default.
- b. **False** Variables declared in the body of a particular member function are local variables. They can only be used within that specific function, not in all member functions of the class.
- c. **False** Empty parentheses following a function name in a function prototype indicate that the function takes no parameters.
- d. **True**

2.

- a. A header file contains class definitions, function prototypes, and constant declarations. Its purpose is to provide an interface for the code, allowing other files to know what functions and classes are available without needing to see the implementation. A source-code file contains the actual implementation of the functions and member functions declared in the header file. Its purpose is to define how the functions perform their tasks.
- b. Set and get functions are used to control access to a class's private data members. This supports the principle of encapsulation. A get function allows external code to read the value of a private data member safely, while a set function allows external code to modify the value, often including validation logic to ensure the data remains valid.

3.

- a. **Error:** The pointer `pS` is uninitialized when it is dereferenced (`*pS`). This leads to undefined behavior. **Correction:** Initialize the pointer to point to a valid `std::string` object before dereferencing it.
- b. **Error:** A pointer of type `float*` cannot be directly assigned to a pointer of type `long*` because they are different data types. **Correction:** Both pointers must be declared with the same data type, such as changing `long* pL` to `float* pL`.

c. **Error:** The statement `std::cout << pX;` prints the memory address stored in pX, not the value of x. **Correction:** Dereference the pointer using the `*` operator to print the value.